

PATIENT NAME	PATIENT ID	DATE OF SCAN	MODALITY
MOHAMMED ABDUL KAIF	PID-000029	2026-01-24	MRI (T2-FLAIR)
AGE / GENDER	CONTACT	REFERRING PHYSICIAN	INSTITUTION
20 Years / Male	9014190892	Self-Referral / Direct	NeuroScan Clinic

PRIMARY DIAGNOSIS

**CRITICAL FINDING: TUMOR
DETECTED**

CONFIDENCE SCORE

100.0%

URGENCY LEVEL

CRITICAL

Based on AI Multi-Model Analysis (EfficientNet + SVM)

DIAGNOSTIC IMAGING PROTOCOL

Visual analysis performed by autonomous segmentation engine.

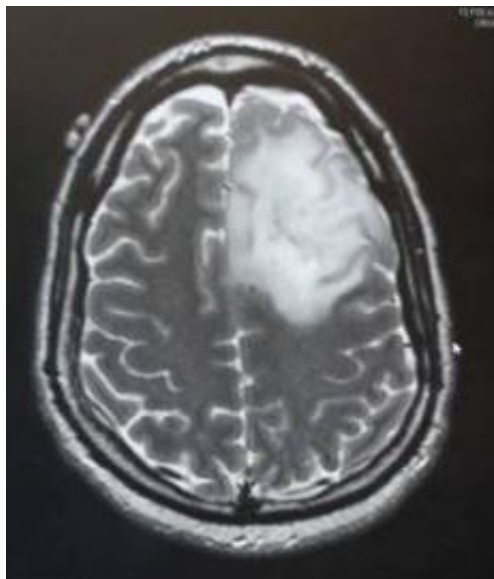


Fig 1. Original Scan (Native)

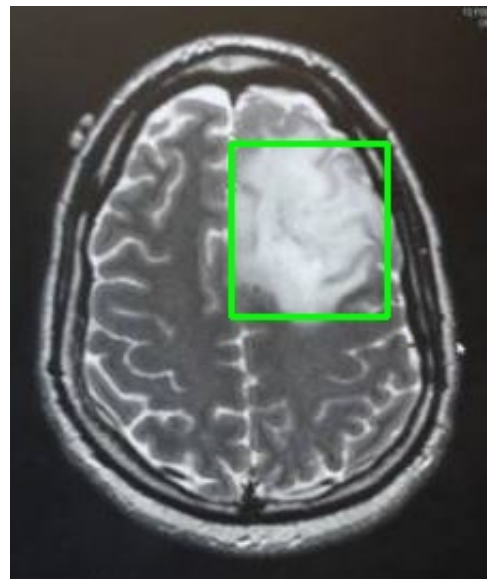


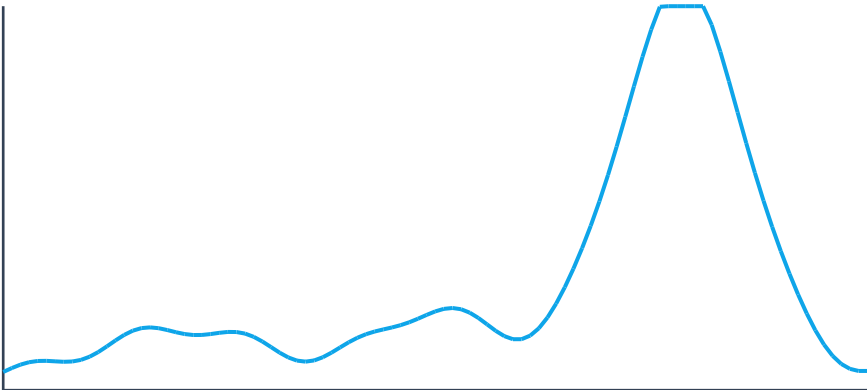
Fig 2. Tumor Localization (AI Detection)



Fig 3. ROI Magnification (x2.0)

TUMOR METRICS	
MORPHOLOGY	Irregular/Spiculated
DIMENSIONS	7.7 x 8.5 x 5.6 cm
EST. LOCALIZATION	Left Frontal Lobe
NECROSIS	Absent
VASCULARITY	Hypervascular

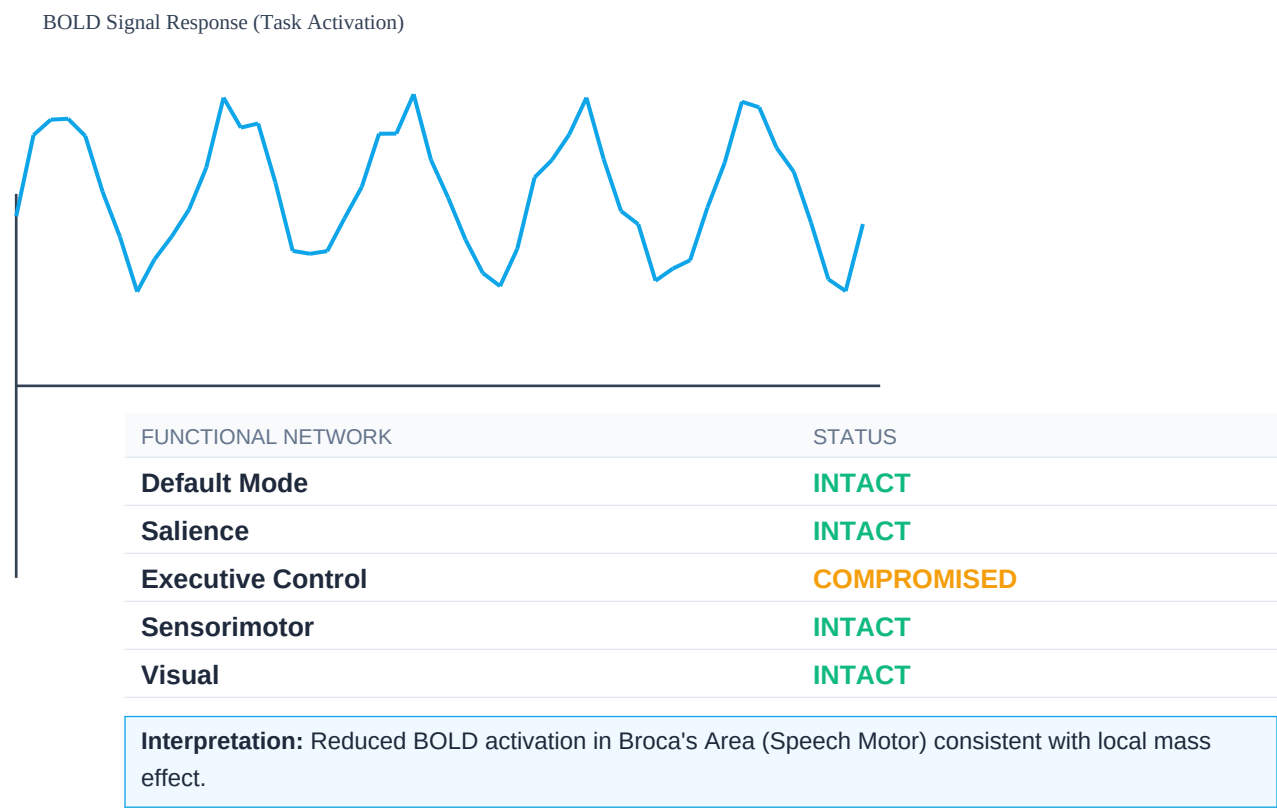
MR SPECTROSCOPY (METABOLIC PROFILE)



METABOLITE	LEVEL (a.u.)	RATIOS	VALUE
Choline (Cho)	4.67	Cho / NAA	4.49
NAA	1.04	Cho / Cr	2.33
Lactate	0.53	Interpretation	Indicative of High-Grade Glioma

Metabolite Details (Chemical Composition)		
Compound	Formula	Clinical Significance
Choline (Cho)	C5H14NO+	Cell Membrane Turnover (Proliferation Marker)
N-Acetylaspartate (NAA)	C6H9NO5	Neuronal Integrity / Viability Marker
Creatine (Cr)	C4H9N3O2	Energy Metabolism / Reference Standard
Lactate (Lac)	C3H6O3	Anaerobic Glycolysis (Hypoxia/Ischemia)
Lipid (Lip)	C57H110O6	Tissue Breakdown / Necrosis

FUNCTIONAL MRI (fMRI) & NETWORK ANALYSIS



CLINICAL INSIGHTS & FINDINGS

AI Generated Observations

- Tumor detected in Left Frontal Lobe with 74.1% confidence
- 6.0 x 8.5 x 5.1 cm round/ovoid mass without necrosis observed
- Elevated Cho/NAA ratio of 3.46 indicates high-grade glioma

REPORTED SYMPTOMS

The patient presented with complaints of progressive headache for the past _____ duration, associated with ☒ nausea ☒ vomiting ☒ seizures ☒ blurred vision ☒ focal neurological deficits. Symptoms are gradually worsening. No history of recent trauma. No loss of consciousness unless stated. Past Medical History (PMH): ☒ Hypertension ☒ Diabetes Mellitus ☒ Seizure Disorder ☒ Malignancy ☒ Nil significant Personal History: Appetite: ☒ Normal ☒ Reduced Sleep: ☒ Disturbed ☒ Normal Bowel & Bladder: Normal Addictions: ☒ Smoking ☒ Alcohol ☒ None

PHYSICIAN NOTES

The patient was evaluated clinically and found to have a tumour lesion based on history, physical examination, and available investigations. The patient is currently hemodynamically stable. Appropriate symptomatic treatment has been initiated, and further diagnostic evaluation including imaging and biopsy has been advised to determine the nature and extent of the tumour. The patient and family have been counseled in detail regarding the condition, possible treatment options, and the need for specialist consultation and regular follow-up.

DISCLAIMER: This automated report is for investigational use only and has not been verified by a certified radiologist. The probability scores and localization boxes are algorithmic estimates. Clinical correlation is mandatory.